```
#All relevant imports for the notebook
import pandas as pd
import matplotlib.pyplot as plt
import requests
import datetime
from datetime import datetime
import sqlite3
import io
import seaborn as sns
from scipy import stats
!pip install plotly
!pip install pycountry
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Does Alcohol Consumption Increase the Chance of Cancer?

Hypothesis / Analysis Goal

Cancer, a genetic disease caused by uncontrollable cell growth and division, is often linked to poor lifestyle habits. Although research have shown that excessive alcohol consumption can lead to cancer, only 45 per cent of Americans believe alcohol has a significant effect on whether someone develops cancer, according to a 2019 national survey by the American Institute for Cancer Research. Hence, in this project, we will use historical alcohol consumption and cancer occurrences to explore the correlation between alcohol consumption and cancer rates in countries all around the world. This analysis may help people learn better about cancer prevention and the effects of alcohol to their health and daily lives.

Data source identification and exploration

Requirement Analysis. To address this problem statement, we need data about cancer occurences, and the data has to be related to the alcohol history before time of diagnosis. For more accurate analysis, we want to have sufficient data regarding

of diagnosis. For more accurate analysis, we want to have sufficient data regarding cancer patients for all possible alcohol consumption history, including different types of alcohol, as well as covering various nationalities and ages. Since data for individual patients will be too many, it is sufficient to consider the average alcohol consumption per capita for different countries, and compare it with cancer occurences in the countries. This ensures that our analysis covers a wide range of geographical locations for more accurate analysis.

List of considered sources. In the following, we outline several sources we considered and explored and justify the final source selection (marked with an *), in particular based on concepts discussed in class.

- **Alcohol consumption**
  - Alcohol by country across 5 different years (DS1) (Source)
  - Country, alcohol type, religion (DS2) * (Source)
  - Alcohol consumption by GDP per capital (DS3) * (Source)
- **Cancer Occurrences**
  - Cancer wrt Race (DS4) (Source)
  - Cancer probability by age and region (DS5) * (Source)

DS1: Alcohol by country across 5 different years

Description. This dataset is from kaggle, downloaded on 3/19/2025. It contains countries, year and alcohol consumption per capita consumed in litres for people aged 15 and older. The countries can be combined with that of the cancer dataset, providing better information on alcohol consumption in tandem with cancer rates for each country. The following code imports the data into a DataFrame and displays the first few rows, thereby giving an idea of the stored data.

```
# Read in the csv file into a DataFrame
df_ds1 = pd.read_csv('sample_data/ds1.csv')
print(df_ds1)
```

In this dataset, alcohol consumption is by litres of pure alcohol, which may be different across different datasets. This must be taken into account when comparing different types of datasets, which may just use raw amount consumed. For example, a 100 ml alcoholic drink with only 5% pure alcohol content will only result in 5 ml of pure alcohol consumed.

alcohol consumed.

Exploring countries in the dataset

```
# Find distinct countries
country_ds1=pd.concat([df_ds1['Entity']]).unique()
print(country_ds1)

# Find the number of distinct countries
num_countries_ds1 = df_ds1['Entity'].nunique()
print(f"number of distinct countries: {num_countries_ds1}")
```

The granularity of the country data seems sufficient enough for analysis, as we need to cover enough countries such that our analysis can be done on a global scale. We note that this data contains entities which are not countries, such as "Middle East and North Africa (WB)", "South Asia (WB)", "Upper-middle-income countries", etc. This needs to be taken into account if we decide to use this dataset as we would need to remove these rows.

Profiling geographical coverage

```
# Ensure Year is integer first (not float or string)
df_ds1['Year'] = df_ds1['Year'].astype(int)

# Convert 'Year' to datetime (set to Jan 1 of that year)
df_ds1['Year'] = pd.to_datetime(df_ds1['Year'])

# Sort the DataFrame by the 'Date' column
df_ds1_sorted = df_ds1.sort_values('Year')

print(df_ds1_sorted)
```

```
# Number of data per country
country_counts_ds1 = df_ds1.groupby('Entity')['Entity'].count().sort_values(
print(country_counts_ds1)
```

```
# Find number of years of data that each country has
# according to country_counts_ds1
countries_number_of_years_ds1 = country_counts_ds1.head(201)

# Create a histogram
plt.figure(figsize=(12, 6))
plt.bar(countries_number_of_years_ds1.index, countries_number_of_years_ds1.v
```

```
plt.xlabel('Country')
plt.ylabel('Number of years')
plt.title('Number of years of data per country')
plt.xticks(rotation=90)  # Rotate x-axis labels for readability
plt.tight_layout()
plt.show()
```

According to this, the dataset contains alcohol consumption from 2000 to 2018, and the highest number of years of data among the countries is 5, which means that there are only less than or equal to 5 years worth of alcohol consumption data per country. This might not be comprehensive enough, considering that alcohol does not directly cause cancer but is rather a build up of alcohol consumption habit over the years. Hence, a dataset with more years per country is preferred.

Exploration of alcohol consumption of countries

```
# Sort alcohol consumption from highest to lowest across all years
alcohol_count_ds1 = df_ds1[['Entity', 'Year', 'Total alcohol consumption per

print(alcohol_count_ds1)
```

According to this dataset, it seems that there may be an outlier among the data points. 98 litres of pure alcohol per capita is quite high, compared to the average of 5.5 litres globally. Additionally, upon further secondary research, the data seems inaccurate (despite reviews on Kaggle affirming its quality), as certain countries including Libya and Somalia banned alcohol since late 1900s. This means that their alcohol consumption per capita should be near 0, but contradictorily, this dataset claims that Libya has the highest alcohol consumption of 98 litres in 2010.

Data quality considerations

We summarize key quality aspects of DS1 as follows:

- The data provided has a wide range of geographical coverage, covering nearly all the countries in the world
- For our task, the data is not sufficiently complete, given that it only provides at most 5 years of alcohol consumption data per country
- Dataset does not seem very reliable based on sense-checking and secondary research

Conclusions for DS1

We disregard DS1 from further consideration because of reliability issues as well as lack of data for alcohol consumption across different years for each country.

DS2: Alcohol Consumption Based on Alcohol Types

Description. This dataset is from Kaggle, downloaded on 31/1/2025. It contains beer servings, spirit servings, wine servings, total litres of pure alcohol as well as religion in each country. For our analysis, the attribute for religion is not relevant, hence we will not project it into the dataframe for further processing.

```
# Read in the csv file into a DataFrame
df_ds2 = pd.read_csv('sample_data/ds2.csv')
print(df_ds2)
```

Exploring countries in the dataset

```
# Find distinct countries
country_ds2=pd.concat([df_ds2['country']]).unique()
print(country_ds2)

# Find the number of distinct countries
num_countries_ds2 = df_ds2['country'].nunique()
print(f"number of distinct countries: {num_countries_ds2}")
```

This dataset provides a comprehensive enough number of countries, given that it provides data for 193 out of 195 countries in the world.

```
# Select only the relevant columns country, beer_servings, spirit_servings,
df_ds2_selected = df_ds2[['country', 'beer_servings', 'spirit_servings', 'wi
print(df_ds2_selected)
```

We can see that the total litres of pure alcohol seem vastly different from the beer, spirit and wine servings. Hence, taking alcohol consumption based on total litres of pure alcohol does not seem to be a good idea, considering that the sources for this is unclear. However, we can still consider the beer, spirit and wine servings as supplementary data to further investigate if different types of alcohol have different

effects on cancer occurrences. However, this dataset lacks data across the years as it only provides data for consumption of wine, spirits and beers at one point in time.

Profiling types of alcohol consumption

```
min_beer = df_ds2['beer_servings'].min()
max_beer = df_ds2['beer_servings'].max()

min_wine = df_ds2['wine_servings'].min()
max_wine = df_ds2['wine_servings'].max()

min_spirit = df_ds2['spirit_servings'].min()
max_spirit = df_ds2['spirit_servings'].max()

# Create a table using pandas DataFrame
table_data_ds2 = {'Metric': ['Minimum beer consumption', 'Maximum beer consu
              'Value': [min_beer, max_beer, min_wine, max_wine, min_spirit,
table_df_ds2 = pd.DataFrame(table_data_ds2)

# Print the table
print(table_df_ds2)
```

All in all, the maximum consumption for all three alcohol categories seems reasonably close to each other, indicating that there are likely no outlier values.

Data quality considerations

We summarize key quality aspects as follows:

- For our task, the data lacks span across the years as it only provides a snapshot of alcohol consumption per category at a point in time.
- Total litres of pure alcohol does not seem accurate, especially when compared to the amounts of the separate categories.
- Nevertheless, source is useful as it further breaks down the total pure alcohol consumption into smaller categories namely beers, wine and spirits.

Conclusions for DS2

We can continue using DS2 as it provides a unique breakdown of pure alcohol consumption which might create more insights on the effects of alcohol on cancer. Despite it only providing data for one year, we believe that the spread of alcohol consumption across categories in each country will likely not change much

considering the circumstances (for example, a wine-producing country will likely consume the much more wine than spirits or beers, and this fact will remain the same over a long period of time).

DS3: Alcohol consumption by GDP per capita

Description. This dataset was obtained from World Health Organisation (via WorldBank). We downloaded the CSV file from Our World in Data on 13/3/2025. This dataset contains data showing the correlation between alcohol consumption per capita versus GDP per capita in various countries. At first glance, this data is much more comprehensive than DS1, given that it provides data for alcohol consumption in each country every year from 1990 to 2023, albeit some missing data for some countries. Furthermore, this data considers the alcohol consumption per capita for people aged 15 and older, which is more relevant to our research question as we can set the legal drinking age to be around 15 years old globally, and compare this data to cancer occurrences for people aged 15 and above only. Thus, we can ensure that we remove any anomalies of cancer occurrences not relating to alcohol consumption (such as in many cases of childhood cancer).

We retain only the columns that are relevant for our analysis, which are the country names, year, and alcohol consumption per capita.

```
# Select only the relevant columns Entity, Year, Total alcohol consumption p
df_ds3 = pd.read_csv('sample_data/ds3.csv')
df_ds3_selected = df_ds3[['Entity', 'Year', 'Total alcohol consumption per c
print(df_ds3_selected)
```

```
country_ds3 = pd.concat([df_ds3_selected['Entity']]).unique()
print(country_ds3)
```

```
avg_alcohol_ds3 = df_ds3.groupby('Entity')['Total alcohol consumption per ca
avg_alcohol_ds3.columns = ['Entity', 'Total alcohol consumption per capita (
print(avg_alcohol_ds3)
```

There are some missing values, which in a lot of the cases here are due to duplicates of the same countries. Hence, we will drop the countries with NaN total alcohol consumption per capita during the data cleaning process.

```
top_20_alcohol= avg_alcohol_ds3.nlargest(20, 'Total alcohol consumption per
bottom_20_alcohol = avg_alcohol_ds3.nsmallest(20, 'Total alcohol consumption
print(top_20_alcohol)
print(bottom_20_alcohol)
```

```
plt.figure(figsize=(12, 6))
plt.bar(top_20_alcohol['Entity'], top_20_alcohol['Total alcohol consumption
plt.xlabel('Entity')
plt.ylabel('Total alcohol consumption per capita (liters of pure alcohol')
plt.title('Top 20 Countries with highest alcohol consumption')
plt.xticks(rotation=90)  # Rotate x-axis labels for readability
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(12, 6))
plt.bar(bottom_20_alcohol['Entity'], bottom_20_alcohol['Total alcohol consum
plt.xlabel('Entity')
plt.ylabel('Total alcohol consumption per capita (liters of pure alcohol')
plt.title('Top 20 Countries with lowest alcohol consumption')
plt.xticks(rotation=90)  # Rotate x-axis labels for readability
plt.tight_layout()
plt.show()
```

Data quality considerations

- Some outlier in certain countries, but the rest of the data seems accurate by comparing to other previous datasets and secondary research
- Source is comprehensive and provide data for the past 20 years of alcohol consumption

Conclusion for DS3

We will proceed using this dataset as it provides a comprehensive coverage and alcohol consumption trend across the years.

DS4: Cancer Occurrences Based on Race

Description. This dataset was downloaded as a CSV file from kaggle on 31/1/2025. It contains 16 attributes, providing information about individual cancer patients and their personal information, including their age, gender, race, BMI, smoking status, family history, cancer type, cancer stage, tumor size as well as treatment information

family history, cancer type, cancer stage, tumor size as well as treatment information. Although other factors such as age, race, BMI, smoking status and family history might provide interesting insights into their effects on cancer, for the sake of this analysis, we will be dropping columns that are redundant to the research question.

```
# Read in the csv file into a DataFrame
df_ds4 = pd.read_csv('sample_data/ds4.csv')
print(df_ds4)
```

```
# Select only the relevant columns PatientID, Race/Ethnicity, CancerType
df_ds4_selected = df_ds4[['PatientID', 'Race/Ethnicity', 'CancerType']]
print(df_ds4_selected)
```

We notice a large number of 'others' in the Race/Ethnicity column, which could pose a trouble when we try to classify the cancer patient based on their race/ethnicity.

Accuracy considerations

By comparing race/ethnicity to the countries in DS2 and DS3, we would be able to get a rough estimate of the effects of alcohol consumption on cancer by grouping countries in the same region together (eg. grouping asian countries together to be used as comparison to the Asian race/ethnicity in DS4). However, this would be less accurate as different countries from the same region might have vastly different alcohol consumption. For example, in the same region of Asia, a country such as Korea with stronger drinking culture would likely have higher alcohol consumption than a country like Malaysia that is run under the Sharia Law (ie. muslims are banned from purchasing and drinking alcohol). As such, we disregard this dataset for future considerations due to the lack of detail of the data provided.

Data quality considerations

We summarise this dataset's key quality aspects as follows:

- Provides specific type of cancer as well as other lifestyle habits which might affect cancer too
- Does not provide the country of nationality, which makes further analysis difficult

Conclusion for DS4

We chose to not proceed with this dataset due to the possibly large assumption we are making by grouping countries of the same region together based on ethnicity and determining effects of alcohol consumption on cancer based on that.

DS5: Cancer occurences incidences

Description. This dataset is downloaded from the World Health Organisation. It summarises the global instances of cancer in 2022, accounting for the different age distributions by adjusting the crude rate to an assumed uniform age in the Age Standardised Rate (ASR).

```
# Read in the csv file into a DataFrame
df_ds5 = pd.read_csv('sample_data/ds5.csv')
print(df_ds5)
```

```
df_ds5_selected = df_ds5[['Label', 'Number', 'ASR (World)', 'Crude rate', 'C
print(df_ds5_selected)
```

```
top_20_high_asr = df_ds5_selected.nlargest(20, 'ASR (World)')
top_20_low_asr = df_ds5_selected.nsmallest(20, 'ASR (World)')
print(top_20_high_asr)
print(top_20_low_asr)
```

to streamline our research. Considering the large number of countries, we decided to compare the top 20 countries with the highest

```
plt.figure(figsize=(12, 6))
plt.bar(top_20_high_asr['Label'], top_20_high_asr['ASR (World)'])
plt.xlabel('Country')
plt.ylabel('Cancer occurrences (ASR)')
plt.title('Top 20 Countries with highest cancer occurrences (ASR)')
plt.xticks(rotation=90)  # Rotate x-axis labels for readability
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(12, 6))
plt.bar(top_20_low_asr['Label'], top_20_low_asr['ASR (World)'])
plt.xlabel('Country')
plt.ylabel('Cancer occurrences (ASR)')
plt.title('Top 20 Countries with lowest cancer occurrences (ASR)')
plt.xticks(rotation=90)  # Rotate x-axis labels for readability
```

```
plt.tight_layout()
plt.show()
```

From our observation on the top 20 countries with the lowest cancer rates, we can see that the data from Sierra Leone is probably an outlier as the cancer rates there is the lowest, and less than 1/2 of the second lowest cancer occurrences (ASR) in the Republic of Gambia. We conclude that this could be due to poor data collection, and decided to simply drop the data for Sierra Leone in our subsequent exploratory steps as we have sufficient data from other countries for our analysis.

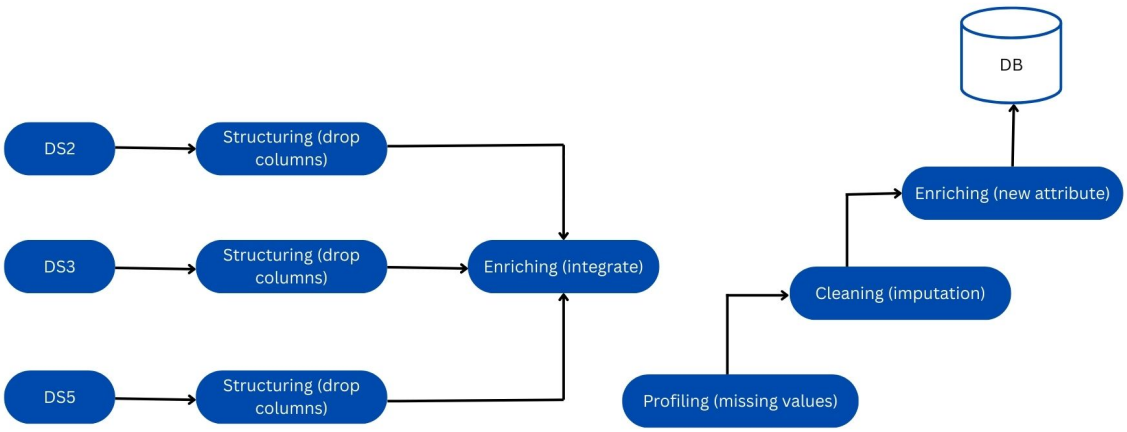
Data quality considerations

- Source seems reliable as it is taken from the World Health Organization data
- Source has a lot of duplicates that needs to be removed later

Conclusion for DS5

We will proceed using DS5 as our data for cancer rates in countries around the world.

Data Preparation (Process)



Overview

To perform data preparation, we implement the following steps: data structuring to obtain a better structured schema, data enriching to integrate our three datasets together, data cleaning to remove erroneous and null values, and lastly data enriching to add data useful for analysis. After profiling for missing values, we perform data structuring, redundant rows or columns, before merging our datasets using inner join. We perform data cleaning, either imputating the missing data with our projected values or removing the rows with too many null values to do imputation. Lastly, we performed data enrichment by defining the level of alcohol consumed based on the three alcohol types: beer, wine and spirit.

Data profiling for missing values

First, we find the number of null values for our datasets, and compile them into a graph to visualise which columns tend to have more null values.

```
df_ds2_selected_null=df_ds2_selected.isnull().sum()
df_ds3_cleaned_null=df_ds3.isnull().sum()
df_ds5_selected_null=df_ds5_selected.isnull().sum()

ds2_nan_percentages = (df_ds2_selected_null / len(df_ds2_selected)) * 100
print(ds2_nan_percentages)
print(df_ds2_selected_null)
ds3_nan_percentages = (df_ds3_cleaned_null / len(df_ds3)) * 100
print(ds3_nan_percentages)
print(df_ds3_cleaned_null)
ds5_nan_percentages = (df_ds5_selected_null / len(df_ds5_selected)) * 100
print(ds5_nan_percentages)
print(df_ds5_selected_null)

plt.figure(figsize=(30, 6))
plt.bar(df_ds2_selected_null.index, df_ds2_selected_null.values, label='df_d
plt.bar(df_ds3_cleaned_null.index, df_ds3_cleaned_null.values, label='df_ds3
plt.bar(df_ds5_selected_null.index, df_ds5_selected_null.values, label='df_d
plt.xlabel('Column')
plt.ylabel('Number of null values')
plt.title('Number of null values per column')
```

Only ds2 contains null values. Hence, we can do an outer join and standardise naming conventions for countries later, without worrying about excessive null values.

Data Structuring

As mentioned earlier, we need to drop Sierra Leone due to its inaccurate data. To facilitate merging later, we standardise the country column name.

```
df_ds2_selected = df_ds2_selected[df_ds2_selected['country'] != 'Sierra Leon
df_ds3 = df_ds3[df_ds3_selected['Entity'] != 'Sierra Leone']
df_ds3 = df_ds3.rename(columns={'Entity': 'country'})
df_ds5_selected_cleaned = df_ds5_selected[df_ds5_selected['Label'] != 'Sierr
df_ds5_selected_cleaned = df_ds5_selected_cleaned.rename(columns={'Label': '



print(df_ds2_selected)
print(df_ds3)
print(df_ds5_selected_cleaned)
```

Because there are multiple repeated countries with different years, we decide to do a pivot table for ds3, facilitating the imputation of null entries later on as well as the merging of datasets.

```
df_ds3_cleaned_pivot = df_ds3.pivot_table(index=['country'], columns='Year',
df_ds3_cleaned_pivot = df_ds3_cleaned_pivot.reset_index()
print(df_ds3_cleaned_pivot)
#
```

Data Enrichment

Now, we can merge all three datasets: ds2, ds3 and ds5. We perform an outer join to minimise loss of data.

```
joined_df=pd.merge(df_ds2_selected, df_ds3_cleaned_pivot, on='country', how=
joined_ds=pd.merge(joined_df, df_ds5_selected_cleaned, on='country', how='ou
print(joined_ds)
print(f"Total null values: {joined_ds.isnull().sum()}")
print(f"Total unique countries left: {joined_ds['country'].nunique()}")
```

Strangely, total unique countries seem to have increased. Upon closer inspection, it seems that certain countries have different names across the datasets, for examples

"Viet Nam" vs "Vietnam". Hence, we will have to clean the data first.

Data cleaning

```
import pycountry
import pandas as pd

# Standard replacements
COUNTRY_REPLACEMENTS = {
    'Antigua & Barbuda': 'Antigua and Barbuda',
    'Bosnia Herzegovina': 'Bosnia and Herzegovina',
    'Bosnia-Herzegovina': 'Bosnia and Herzegovina',
    'Cape Verde': 'Cabo Verde',
    'Congo, Republic of': 'Congo',
    'DR Congo': 'Congo, The Democratic Republic of the',
    'Congo, Democratic Republic of the': 'Congo, The Democratic Republic of
    'Congo, Democratic Republic of': 'Congo, The Democratic Republic of the'
    'Cote d\'Ivoire': 'Côte d\'Ivoire',
    'France (metropolitan)': 'France',
    'France, Guadeloupe': 'France',
    'France, La Réunion': 'France',
    'France, Martinique': 'France',
    'French Guyana': 'France Guiana',
    'French Southern Territories': 'France',
    'USA': 'United States',
    'UK': 'United Kingdom',
    'Republic of Korea': 'Korea, Republic of',
    'South Korea': 'Korea, Republic of',
    'North Korea': 'Korea, Democratic People\'s Republic of',
    'Korea (Democratic People Republic of)': 'Korea, Democratic People\'s Re
    'Czechia': 'Czech Republic',
    'Vietnam': 'Viet Nam',
    'Iran': 'Iran, Islamic Republic of',
    'Iran, (Islamic Republic of)': 'Iran, Islamic Republic of',
    'Sao Tome & Principe': 'Sao Tome and Principe',
    'Syria': 'Syrian Arab Republic',
    'Trinidad & Tobago': 'Trinidad and Tobago',
    'Turkey': 'Türkiye',
    'Taiwan': 'Taiwan, Province of China',
    'Bolivia': 'Bolivia, Plurinational State of',
    'Tanzania': 'Tanzania, United Republic of',
    'Venezuela': 'Venezuela, Bolivarian Republic of',
    'Laos': 'Lao People\'s Democratic Republic',
    'Brunei': 'Brunei Darussalam',
    'Moldova': 'Moldova, Republic of',
    'Russia':'Russian Federation'
    }
```

```
def standardize_country_name(name):
    if pd.isna(name):
        return None
    name = str(name).strip()
    name = COUNTRY_REPLACEMENTS.get(name, name)
    try:
        return pycountry.countries.lookup(name).name
    except LookupError:
        return name

def standardize_country_column(df, column_name):
    df['country'] = df['country'].apply(standardize_country_name)
    return df

def convert_iso_numeric_to_name(code):
    if pd.isna(code):
        return None
    try:
        code_int = int(code)
        for country in pycountry.countries:
            if hasattr(country, 'numeric') and int(country.numeric) == code_
                return country.name
    except (ValueError, TypeError):
        return None
    return None

# Apply transformations
df_ds2_selected = standardize_country_column(df_ds2_selected, 'country')
df_ds3_cleaned_pivot = standardize_country_column(df_ds3_cleaned_pivot, 'cou
df_ds5_selected_cleaned = standardize_country_column(df_ds5_selected_cleaned
```

Data enriching by integrating DS2, DS3 and DS5

```
# Inner join
joined_ds = pd.merge(df_ds2_selected, df_ds3_cleaned_pivot, on='country', ho
joined_ds = pd.merge(joined_ds, df_ds5_selected_cleaned, on='country', how='

# Summary
print(joined_ds.head())
print(f"\nTotal null values:\n{joined_ds.isnull().sum()}")
print(f"\nTotal unique countries left: {joined_ds['country'].nunique()}")
```

After further cleaning and integrating the datasets, we are left with 166 countries, which is a decently comprehensive data out of all the 195 countries in the world, and

would likely be sufficient to analyse our problem statement.

Data imputation for merged dataset

```
# Null values in the merged dataset
print(joined_ds[joined_ds.isnull().any(axis=1)])
```

We notice that Montenegro and Serbia have quite a bit of null values between 2000 to 2020, with only data for alcohol consumption for 2010, 2015 and 2019. For data imputation, we decided to take an average of these three countries as the alcohol consumption for the rest of the years, considering the fact that their alcohol consumption remain generally in the similar range and does not deviate too much between these years.

```
import pandas as pd
for column in joined_ds.columns:
    if joined_ds[column].dtype in ['float64', 'int64']:  # Check if column i
        joined_ds[column] = joined_ds[column].fillna(joined_ds[column].media
print(joined_ds[joined_ds.isnull().any(axis=1)])
```

Structuring by dropping total_litres_of_pure_alcohol column

Since we have sufficient data from ds3 for the total alcohol consumption, we will drop the total_litres_of_pure_alcohol column.

```
# remove total_litres_of_pure_alcohol column
joined_ds = joined_ds.drop(columns=['total_litres_of_pure_alcohol'])
print(joined_ds)

pd.set_option('display.max_rows', None)
```

Data enriching by adding new attributes (beer, wine and spirit consumption level)

We categorise countries' consumption level of beer, wine and spirit into low, medium, high and ultra high and analysed the differenes in consumption between countries.

```
import pandas as pd

# Assuming joined_ds is already defined and contains beer_servings, w

# Calculate percentiles for each column
beer_low = joined_ds['beer_servings'].quantile(0.3)
beer_high = joined_ds['beer_servings'].quantile(0.6)
ultra_high = joined_ds['beer_servings'].quantile(0.9)

wine_low = joined_ds['wine_servings'].quantile(0.3)
wine_high = joined_ds['wine_servings'].quantile(0.6)
ultra_high = joined_ds['wine_servings'].quantile(0.9)

spirit_low = joined_ds['spirit_servings'].quantile(0.3)
spirit_high = joined_ds['spirit_servings'].quantile(0.6)
ultra_high = joined_ds['spirit_servings'].quantile(0.9)

def apply_servings_level_dynamic(servings, low, high):
    if servings < low:
        return 'Low'
    elif low <= servings < high:
        return 'Medium'
    elif high<= servings <ultra_high:
        return 'High'
    elif servings>= ultra_high:
        return 'Ultra High'
    else:
        return 'Unknown'

joined_ds['beer_consumption_level'] = joined_ds['beer_servings'].appl
joined_ds['wine_consumption_level'] = joined_ds['wine_servings'].appl
joined_ds['spirit_consumption_level'] = joined_ds['spirit_servings'].

total = len(joined_ds)

print(f"Percentage of beer_consumption_level: "
      f"Low: {((joined_ds['beer_consumption_level'] == 'Low').sum() /
      f"Medium: {((joined_ds['beer_consumption_level'] == 'Medium').s
      f"High: {((joined_ds['spirit_consumption_level'] == 'High').sum
      f"Ultra_High: {((joined_ds['spirit_consumption_level'] == 'Ultr

print(f"Percentage of wine_consumption_level: "
      f"Low: {((joined_ds['wine_consumption_level'] == 'Low').sum() /
      f"Medium: {((joined_ds['wine_consumption_level'] == 'Medium').s
      f"High: {((joined_ds['spirit_consumption_level'] == 'High').sum
      f"Ultra_High: {((joined_ds['spirit_consumption_level'] == 'Ultr
```

```
print(f"Percentage of spirit_consumption_level: "
      f"Low: {((joined_ds['spirit_consumption_level'] == 'Low').sum()
      f"Medium: {((joined_ds['spirit_consumption_level'] == 'Medium')
      f"High: {((joined_ds['spirit_consumption_level'] == 'High').sum
      f"Ultra_High: {((joined_ds['spirit_consumption_level'] == 'Ultr
```

Histogram for beer, wine, and spirit consumption level

```
import matplotlib.pyplot as plt
import seaborn as sns

def plot_servings_distribution_same_range(df, beer_low, beer_high, ul
    min_value = min(df['beer_servings'].min(), df['wine_servings'].mi
    max_value = max(df['beer_servings'].max(), df['wine_servings'].ma

    fig, axes = plt.subplots(3, 1, figsize=(10, 18))

    sns.histplot(df['beer_servings'], bins=20, kde=True, color='skybl
    axes[0].axvline(x=beer_low, color='red', linestyle='--', label=f'
    axes[0].axvline(x=beer_high, color='orange', linestyle='--', labe
    axes[0].axvline(x=ultra_high, color='green', linestyle='--', labe
    axes[0].set_title('Beer Servings Distribution with Consumption Le
    axes[0].set_xlabel('Beer Servings')
    axes[0].set_ylabel('Density')
    axes[0].legend()
    axes[0].set_xlim(min_value, max_value)

    sns.histplot(df['wine_servings'], bins=20, kde=True, color='orang
    axes[1].axvline(x=wine_low, color='purple', linestyle='--', label
    axes[1].axvline(x=wine_high, color='brown', linestyle='--', label
    axes[1].axvline(x=ultra_high, color='green', linestyle='--', labe
    axes[1].set_title('Wine Servings Distribution with Consumption Le
    axes[1].set_xlabel('Wine Servings')
    axes[1].set_ylabel('Density')
    axes[1].legend()
    axes[1].set_xlim(min_value, max_value)

    sns.histplot(df['spirit_servings'], bins=20, kde=True, color='gre
    axes[2].axvline(x=spirit_low, color='blue', linestyle='--', label
    axes[2].axvline(x=spirit_high, color='pink', linestyle='--', labe
    axes[2].axvline(x=ultra_high, color='green', linestyle='--', labe
    axes[2].set_title('Spirit Servings Distribution with Consumption
    axes[2].set_xlabel('Spirit Servings')
    axes[2].set_ylabel('Density')
    axes[2].legend()
    axes[2].set_xlim(min_value, max_value)

    plt.tight_layout()
```

```
    plt.show()

plot_servings_distribution_same_range(joined_ds, beer_low, beer_high,
```

Data analysis (analyse)

In order to answer our research question, we perform the following analysis:

- Descriptive analysis: Visualisation
- Diagnostic analysis: Linear regression analysis

Visualisation

Cloropleth map for alcohol consumption level by country, per category

```
level_mapping = {
    'Low': 1,
    'Medium': 2,
    'High': 3,
    'Ultra High': 4
}

joined_ds['beer_level_num'] = joined_ds['beer_consumption_level'].map(level_
joined_ds['wine_level_num'] = joined_ds['wine_consumption_level'].map(level_
joined_ds['spirit_level_num'] = joined_ds['spirit_consumption_level'].map(le
```

```
import plotly.express as px

fig = px.choropleth(
    joined_ds,
    locations="country",
    locationmode="country names",
    color="beer_level_num",
    hover_name="country",
    color_continuous_scale="YlOrRd",
    title="Beer Consumption Level by Country"
)

fig.show()
```

We can see that regions like North America, majority of South America and Europe, as

We can see that regions like North America, majority of South America and Europe, as well as Australia and Russia has ultra high beer consumption level. Meanwhile, most of Africa and China seem to hvae relatively lower beer consumption, with very low amounts in most African countries excluding some South African countries.

```
import plotly.express as px

fig = px.choropleth(
    joined_ds,
    locations="country",
    locationmode="country names",
    color="wine_level_num",
    hover_name="country",
    color_continuous_scale="YlOrRd",
    title="Wine Consumption Level by Country"
)

fig.show()
```

As compared to beer, the consumption of wine is generally lower worldwide, with only a small part of South America, Europe and Australia having ultra high consumption level.

```
import plotly.express as px

fig = px.choropleth(
    joined_ds,
    locations="country",
    locationmode="country names",
    color="spirit_level_num",
    hover_name="country",
    color_continuous_scale="YlOrRd",
    title="Spirit Consumption Level by Country"
)

fig.show()
```

Consumption of spirits is particularly high for Russia, China and Thailand, while it is unusually lower for Australia (as compared to beer and wine). Overall trend remains similar, with Africa and Asia generally having lower consumption level (excluding China and Russia).

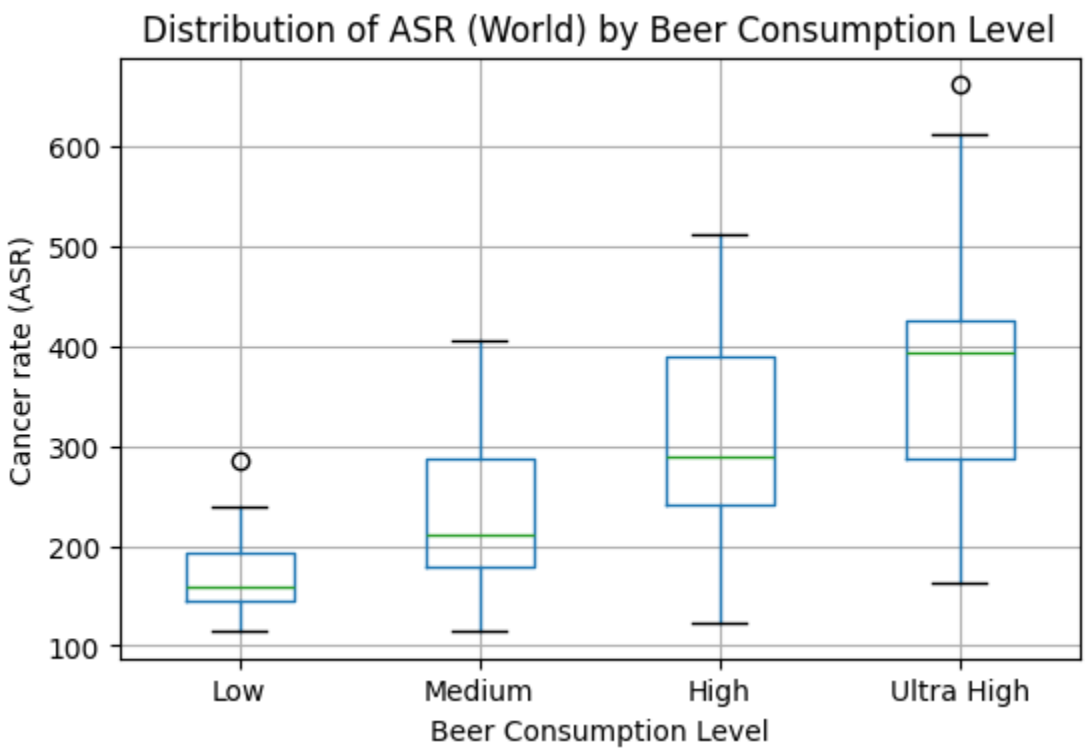
Distribution of cancer rates by alcohol consumption level

Goal: The goal is to visually describe the distribution of cancer rates (ASR (World)) by consumption level of different alcohol types.

Methods / measures: We use box plots as compact visualization of each alcohol type distribution.

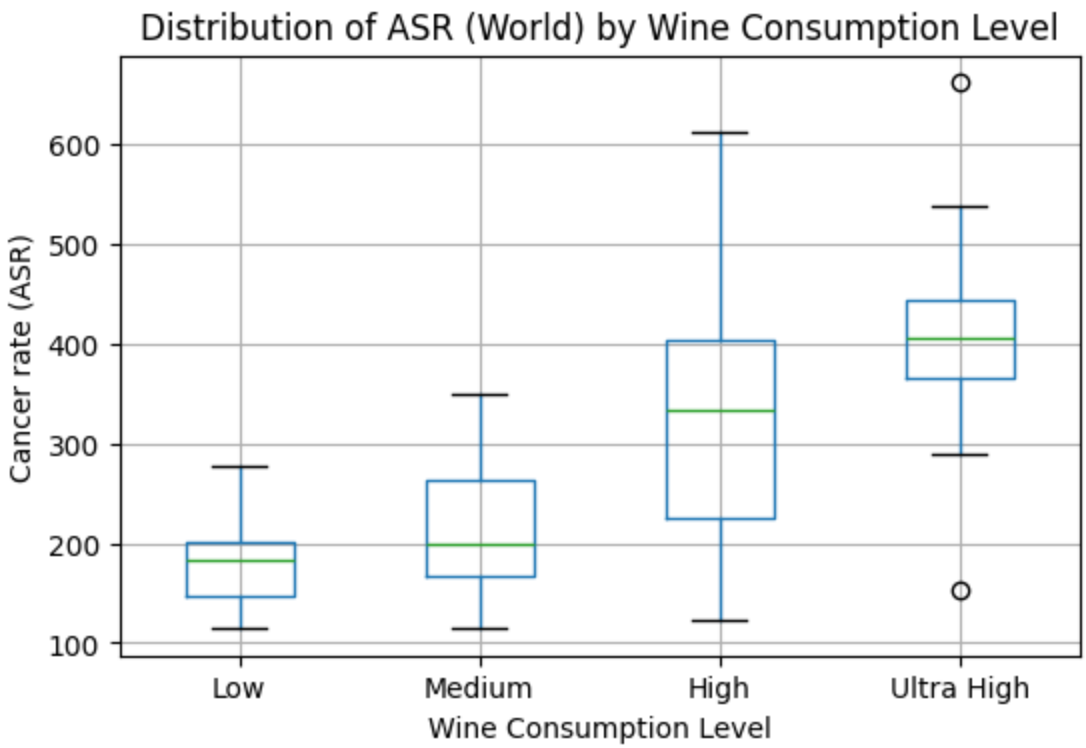
Code: For each climate zone, we generate a box plot.

```
# Boxplot to compare beer consumption level to cancer rates

ordered_levels = ['Low', 'Medium', 'High', 'Ultra High']
joined_ds['beer_consumption_level'] = pd.Categorical(
    joined_ds['beer_consumption_level'],
    categories=ordered_levels,
    ordered=True
)

joined_ds.boxplot(column='ASR (World)', by='beer_consumption_level', figsize
plt.title('Distribution of ASR (World) by Beer Consumption Level')
plt.suptitle('')  # Remove the default suptitle
plt.ylabel('Cancer rate (ASR)')
plt.xlabel('Beer Consumption Level')
plt.show()
```



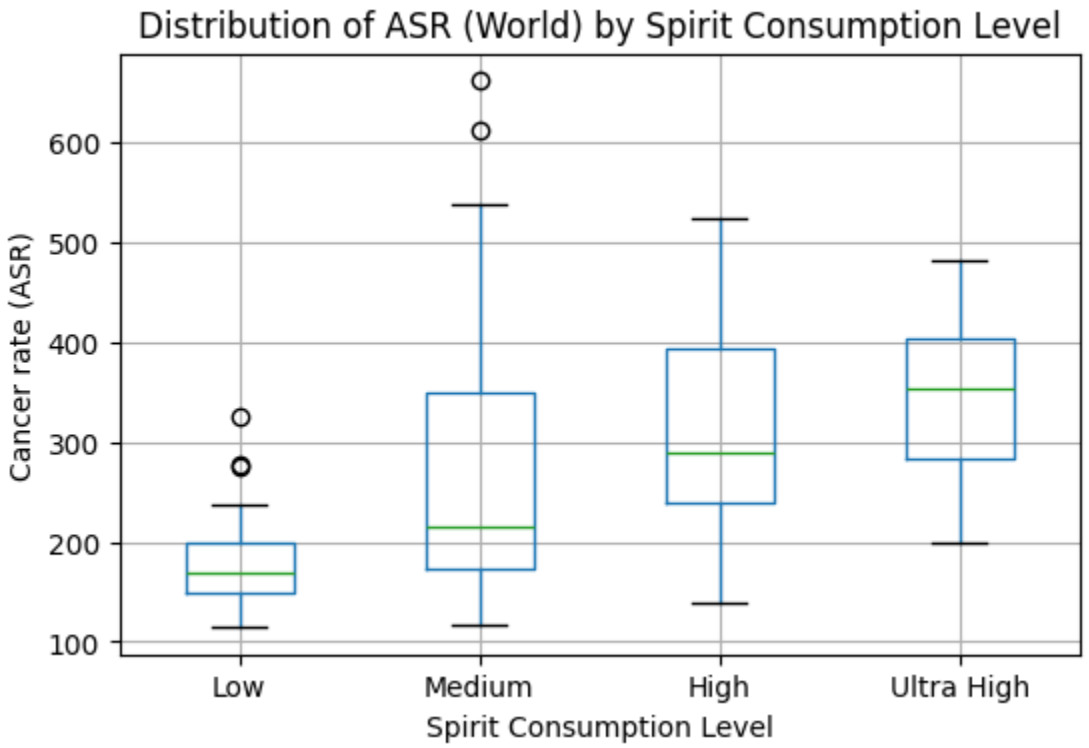

```
# Boxplot to compare wine consumption level to cancer rates

ordered_levels = ['Low', 'Medium', 'High', 'Ultra High']
```

```
joined_ds['wine_consumption_level'] = pd.Categorical(
    joined_ds['wine_consumption_level'],
    categories=ordered_levels,
    ordered=True
)

joined_ds.boxplot(column='ASR (World)', by='wine_consumption_level', figsize
plt.title('Distribution of ASR (World) by Wine Consumption Level')
plt.suptitle('')  # Remove the default suptitle
plt.ylabel('Cancer rate (ASR)')
plt.xlabel('Wine Consumption Level')
plt.show()
```




```
# Boxplot to compare spirit consumption level to cancer rates


ordered_levels = ['Low', 'Medium', 'High', 'Ultra High']
joined_ds['spirit_consumption_level'] = pd.Categorical(
    joined_ds['spirit_consumption_level'],
    categories=ordered_levels,
    ordered=True
)

joined_ds.boxplot(column='ASR (World)', by='spirit_consumption_level', figsi
plt.title('Distribution of ASR (World) by Spirit Consumption Level')
plt.suptitle('')  # Remove the default suptitle
plt.ylabel('Cancer rate (ASR)')
plt.xlabel('Spirit Consumption Level')
```

```
plt.show()
```



Interpretation. The boxplot visualised a similar distribution for all three types of alcohol. From all three types of alcohol, we can see a general pattern of increasing cancer rates as consumption level of alcohol increases from low, to medium, to high, to ultra high. Initially, there is an increasing trend for both the cancer rate (ASR(world)) and the alcohol type consumption level. However, interestingly, the difference between high and ultra high consumption level's effect on cancer rate is not as striking as the lower end of consumption levels. This might indicate that there is stronger correlation between alcohol consumption to cancer rates when comparing lower amounts of alcohol.

Limitations. We acknowledge that our boundary on the definition of alcohol consumption level for each type of alcohol is purely based on statistical value and not based on any other backed research/source. Hence, concluding general correlation based on this data would be possible but any specific analysis (such as regarding the correlation strength of different levels of consumption would require follow-up statistical analysis (such as hypothesis testing below).

Relationship and dependency analysis

Goal: Identify if there is a correlation between consumption of alcohol and cancer

rates after 5 years, and whether the specific type of alcohol consumed impacts the cancer rate.

Method: Implement a Pearson correlation analysis between the alcohol factors (beer/wine/alcohol consumption amount, total amount of pure alcohol consumed) and cancer factors (ASR(world), crude rate, cumulative risk)

Code: Compute the Pearson correlation constant as well as the p-values for both, between the two groups of factors described above, as well as do a summary of their correlation statistics.

```
#Pearson correlation
year_cols=[int(year) for year in range(2000, 2021)]
joined_ds['average_alcohol_consumed']=joined_ds[year_cols].mean(axis=1)
alcohol_summary = ['beer_servings', 'wine_servings', 'average_alcohol_consum
cancer_summary = ['ASR (World)']

for i in alcohol_summary:
  for j in cancer_summary:
    correlation, p_value = stats.pearsonr(joined_ds[i], joined_ds[j])
    print(f"Pearson correlation between {i} and {j}: {correlation}, p-value:


import statsmodels.api as sm
import pandas as pd

X = joined_ds[['beer_servings', 'wine_servings', 'spirit_servings', 'average
X = sm.add_constant(X)

y = joined_ds[['average_alcohol_consumed']]

model = sm.OLS(y, X).fit()

summary = model.summary2().tables[1]

def add_significance_stars(p_value):
    if p_value < 0.0005:
        return '***'
    elif p_value < 0.005:
        return '**'
    elif p_value < 0.05:
        return '*'
    else:
        return ''

summary['Significance'] = summary['P>|t|'].apply(add_significance_stars)

print(summary[['Coef.', 'P>|t|', 'Significance']])
```

```
Pearson correlation between beer_servings and ASR (World): 0.6639629989752889
Pearson correlation between wine_servings and ASR (World): 0.6662077642233368
Pearson correlation between average_alcohol_consumption and ASR (World): 0.69
                                Coef.         P>|t| Significance
const                     4.213602e-16  3.817220e-01
beer_servings             1.521057e-17  3.087873e-04          ***
wine_servings            -3.131546e-17  3.266643e-12          ***
spirit_servings           2.013586e-17  4.939539e-06          ***
average_alcohol_consumed  1.000000e+00  0.000000e+00          ***
```

Interpretation. Based on this correlation analysis, we can see that there is a positive correlation between level of alcohol consumption and cancer rates (ASR (World)) for all three types of alcohol. Although they do not show a very strong correlation, we can see that their p-value is very low, meaning that this corrrelation is very statistically significant.

Limitation. We did not consider the trend of alcohol consumption over the years as we took an average out of it. This might be a factor that can affect the correlation further, especially if there are recent or gradual changes to alcohol consumption level in the country over the years.

Presentation of main results (share)

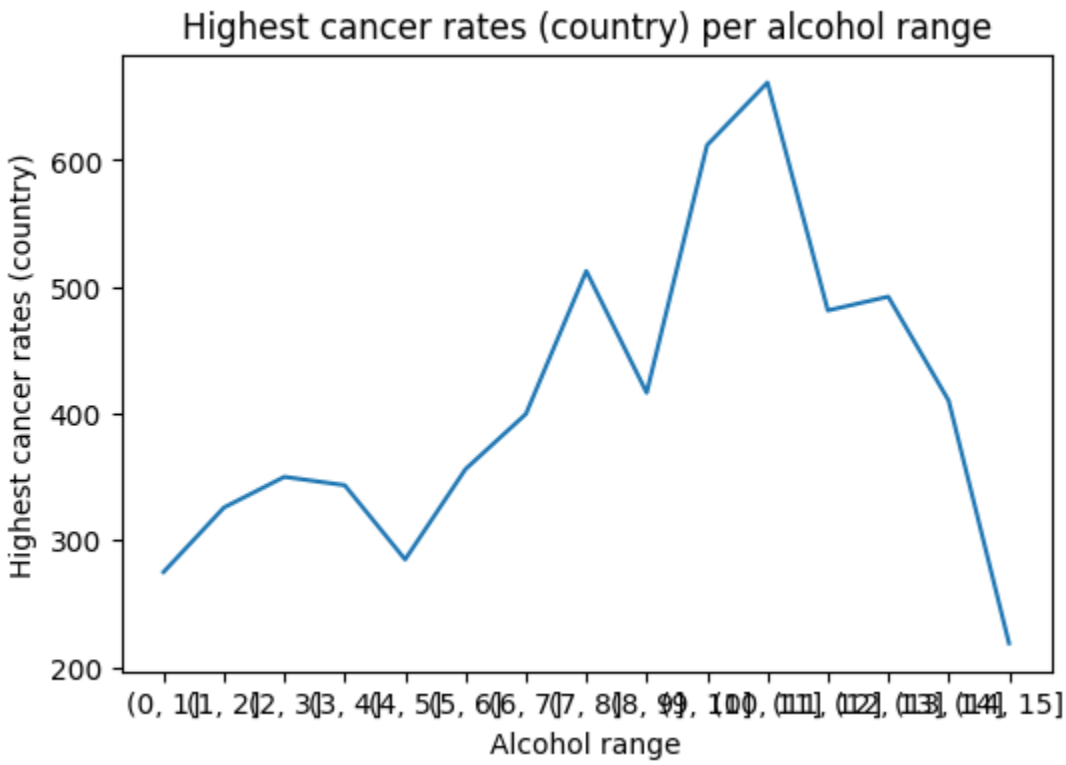
Stakeholders. Our results can be presented to the public, particularly people who consume alcohol.

Visualisation

```
# Create alcohol consumption range
joined_ds['alcohol_range'] = pd.cut(joined_ds['average_alcohol_consumed'], b

# Group by alcohol consumption range and get the maximum cancer rates (ASR)
max_cancer_per_range = joined_ds.groupby('alcohol_range')['ASR (World)'].max

# Create the line plot
plt.figure(figsize=(6, 4))
plt.plot(max_cancer_per_range.index.astype(str), max_cancer_per_range.values
```

```
plt.xlabel('Alcohol range')
plt.ylabel('Highest cancer rates (country)')
plt.title('Highest cancer rates (country) per alcohol range')
plt.show()
```

```
<ipython-input-1136-32bc2248df7d>:5: FutureWarning:

The default of observed=False is deprecated and will be changed to True in a
```



Narrative. Our detailed analysis of the effects of cancer rates depending on alcohol consumption has revealed an interesting result. The graphs show the discovered trend, where on lower alcohol consumption, small changes in alcohol consumption can lead to high increase in cancer rates. However, at higher alcohol consumption, the effect on cancer does not seem to be that great. This is similar to our initial visual observation of the box plot comparing the effect of different alcohol types on cancer rates.

Possible actions based on results (Act)

Based on these findings, health organisations should further disencourage the consumption of alcohol to minimise cancer occurences among the general public. As it stands, the data corroborates the fact that even low consumption of alcohol has a

probable likelihood of increasing cancer rates.

Final Reflection

Considerations on data ethics

We did not give credit to the relevant sources of the data as well as the original authors of the datasets. As such, we must remember to cite the sources before publishing.

Plus, we need to make sure the use of data is not in a manner that perpetuates harmful stereotypes, especially concerning such sensitive data like alcohol consumption and cancer risk.

Considerations on data privacy and security

These datasets are made by accessing worldwide patient data. It is unknown whether the creators of the datasets have gotten consent from the cancer patients to access their sensitive data. However, given that the datasets compile data into worldwide, public data instead of individual personal data, we can assume that this issue is of minimal concern.